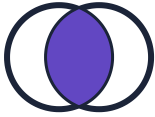




SQL JOIN cheat sheet

LearnHub course reference: Relational Database Design with SQL

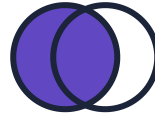
The examples use two tables: **Students**(Id, FullName) and **Enrolments**(Id, StudentId, CourseId). The shaded area in each diagram shows which rows are returned.



INNER JOIN

Only students who have at least one enrolment.

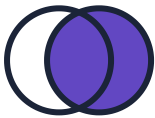
```
SELECT s.FullName,  
       e.CourseId  
FROM Students AS s  
INNER JOIN Enrolments AS e  
      ON e.StudentId = s.Id;
```



LEFT JOIN

Every student; enrolment columns are NULL when there is no match.

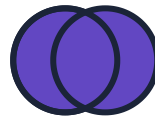
```
SELECT s.FullName,  
       e.CourseId  
FROM Students AS s  
LEFT JOIN Enrolments AS e  
      ON e.StudentId = s.Id;
```



RIGHT JOIN

Every row from the right table. Usually rewritten as a LEFT JOIN with the tables swapped.

```
SELECT s.FullName,  
       e.CourseId  
FROM Students AS s  
RIGHT JOIN Enrolments AS e  
      ON e.StudentId = s.Id;
```



FULL OUTER JOIN

All rows from both tables, matched where possible. Supported by SQL Server and SQLite 3.39+.

```
SELECT s.FullName,  
       e.CourseId  
FROM Students AS s  
FULL OUTER JOIN Enrolments  
      AS e  
      ON e.StudentId = s.Id;
```

Find rows without a match

```
-- Students who have not enrolled in any  
course  
SELECT s.FullName  
FROM Students AS s  
LEFT JOIN Enrolments AS e ON e.StudentId =  
s.Id  
WHERE e.Id IS NULL;
```

Summarise with GROUP BY and HAVING

```
SELECT c.Title, COUNT(e.Id) AS Learners  
FROM Courses AS c  
LEFT JOIN Enrolments AS e ON e.CourseId =  
c.Id  
GROUP BY c.Title  
HAVING COUNT(e.Id) >= 5  
ORDER BY Learners DESC;
```

Clause order

Written	Runs	Purpose
SELECT	5	Choose output columns
FROM / JOIN	1	Combine the tables
WHERE	2	Filter rows
GROUP BY	3	Form groups
HAVING	4	Filter groups
ORDER BY	6	Sort the result

Tips

- Always join on key columns and give tables short aliases.
- Put conditions on the right-hand table of a LEFT JOIN in the ON clause, or unmatched rows

disappear.

- Index foreign key columns used in joins.

LearnHub course reference. Examples use standard SQL that runs on SQL Server and SQLite.